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Satellite Earth Stations and Systems (SES);
Deployment scenarios and
hENs merger analysis

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ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Satellite Earth Stations and Systems (SES).

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Executive summary

The present document reviews the contents of several Harmonised Standards under the purview of ETSI TC SES to ensure their requirements remain current amid the rapid expansion of satellite-based services. The study contained in the present document is focused on an initial list provided by the contractual requirements of ETSI plus additional ENs proposed by experts, in summary, assessing the hEN under review for continued compliance with the RED while identifying specific topics requiring further consideration due to technology evolution since the ENs were initially produced. The study contained in the present document focuses on identifying the scenarios and use cases where the targeted hENs are used and finally the present document provides a prioritized list of hEN to be reviewed in detail.

1 Scope

The present document, as part of the Experts Group, aims to deliver a report covering deployment scenarios, Mapping hENs and use cases resulting from the recommendations and analysis reported in the ETSI TR 104 305 [i.1]

The present document analyses new functionalities and enhancements required for each service (FSS with LEO, FSS with MEO, FSS with GEO, MSS with LEO, MSS with GEO, etc.), based on the use cases and requirements identified in ETSI TR 104 305 [i.1]. It proposes various solutions that can support these enhancements. The solutions discussed include deployment scenarios for each use case. The present document also prioritizes the group of hENs which will require urgent technology updates.

In the analysis of the study contained in the present document, due examination has been taken also on potential for merger some existing ENs into single a (i.e. several hENs addressing earth stations in motion). However, the merger of several hENs, where these hENs are already included in a variety of regulatory frameworks (at international and national levels) raises significant challenges because while the technical merger of some hENs will represent a major improvement and efficiency of the hENs from the technical point of view, such merger would have the potential to raise additional work in the legal and regulatory frames, where those hENs are already mentioned or adopted. The inclusion of new hENs in the standardization system of each country follows specific national procedures the timing of which may raise issues beyond the technical advantages of a potential merger of some hENs.

A prioritization of hENs is also carried out.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 104 305 (V1.1.1): "Satellite Earth Stations and Systems (SES); Study of potential updates to certain harmonised ENs due to technology evolution, methods of measurements and regulatory updates".

[i.2] ETSI EN 303 699: "Satellite Earth Stations and Systems (SES); Fixed earth stations communicating with non-geostationary satellite systems in the 20 GHz and 30 GHz FSS bands; Harmonised Standard for access to radio spectrum".

[i.3] ETSI EN 303 978: "Satellite Earth Stations and Systems (SES); Earth Stations on Mobile Platforms (ESOMP) communicating with satellites in geostationary orbit, operating in the 27,5 GHz to 30,0 GHz and 17,3 GHz to 20,2 GHz frequency bands; Harmonised Standard for access to radio spectrum".

[i.4] ETSI EN 303 979: "Satellite Earth Stations and Systems (SES); Harmonised Standard for Earth Stations on Mobile Platforms (ESOMP) transmitting towards satellites in non-geostationary orbit, operating in the 27,5 GHz to 29,1 GHz and 29,5 GHz to 30,0 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.5] ETSI EN 303 980: "Satellite Earth Stations and Systems (SES); Fixed and in-motion Earth Stations communicating with non-geostationary satellite systems (NEST) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum".

[i.6] ETSI EN 301 427: "Satellite Earth Stations and Systems (SES); Harmonised Standard for low data rate Mobile satellite Earth Stations (MES) except aeronautical mobile satellite earth stations, operating in the 11/12/14 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.7] ETSI EN 301 442: "Satellite Earth Stations and Systems (SES); Harmonised Standard for NGSO Mobile Earth Stations (MES) including handheld earth stations, for Satellite Personal Communications Networks (S-PCN) operating in the 1 980 MHz to 2 010 MHz (earth-to-space) and 2 170 MHz to 2 200 MHz (space-to-earth) frequency bands under the Mobile Satellite Service (MSS)".

[i.8] ETSI EN 301 443: "Satellite Earth Stations and Systems (SES); Harmonised Standard for Very Small Aperture Terminal (VSAT); Transmit-only, transmit-and-receive, receive-only satellite earth stations operating in the 4 GHz and 6 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.9] ETSI EN 301 447: "Satellite Earth Stations and Systems (SES); Satellite Earth Stations on board Vessels (ESVs) operating in the 4/6 GHz frequency bands allocated to the Fixed Satellite Service (FSS); Harmonised Standard for access to radio spectrum".

[i.10] ETSI EN 301 459: "Satellite Earth Stations and Systems (SES); Harmonised Standard for Satellite Interactive Terminals (SIT) and Satellite User Terminals (SUT) transmitting towards satellites in geostationary orbit, operating in the 29,5 GHz to 30,0 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.11] ETSI EN 301 721: "Satellite Earth Stations and Systems (SES); Harmonised Standard for Mobile Earth Stations (MES) providing Low Bit Rate Data Communications (LBRDC) using Low Earth Orbiting (LEO) satellites operating below 1 GHz frequency band".

[i.12] ETSI EN 302 340: "Satellite Earth Stations and Systems (SES); Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 11/12/14 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.13] ETSI EN 302 448: "Satellite Earth Stations and Systems (SES); Harmonised Standard for tracking Earth Stations on Trains (ESTs) operating in the 14/12 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.14] ETSI EN 302 574 (all parts): "Satellite Earth Stations and Systems (SES); Harmonised Standard for Mobile Earth Stations (MES) operating in the 1 980 MHz to 2 010 MHz (earth-to-space) and 2 170 MHz to 2 200 MHz (space-to-earth) frequency bands; Part 3: User Equipment (UE) for narrowband systems".

[i.15] ETSI EN 303 984: "Satellite Earth Stations and Systems (SES); Earth Stations on board Aircraft (ESOA) communicating with non-geostationary satellite systems in the 10,7 GHz to 12,75 GHz and 12,75 GHz to 13,25 GHz frequency bands; Harmonised Standard for access to radio spectrum".

[i.16] ETSI EN 302 977: "Satellite Earth Stations and Systems (SES); Harmonised Standard for Vehicle-Mounted Earth Stations (VMES) operating in the 14/12 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.17] ETSI EN 303 981: "Satellite Earth Stations and Systems (SES); Fixed and in-motion Wide Band Earth Stations communicating with non-geostationary satellite systems (WBES) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum".

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AES	Aeronautical Earth Station
CMF	Control and Monitoring Functions
EIRP	Equivalent Isotropically Radiated Power
EN	European Standard
ESIM	Earth Station In Motion
ESOA	Earth Station On board Aircraft
ESOMP	Earth Station On Mobile Platform
EST	Earth Station on Train
ESV	Earth Station on Vessel
ETSI	European Telecommunications Standards Institute
EU	European Union
FSS	Fixed Satellite Service
GEO	Geostationary Earth Orbit
GSO	Geostationary Satellite Orbit
hEN	Harmonised Standard
IoT	Internet of Things
LBRDC	Low Bit Rate Data Communications
LEO	Low Earth Orbit
MEO	Medium Earth Orbit
MES	Mobile Earth Station
MSS	Mobile Satellite Service
NEST	Fixed and in-motion Earth Stations communicating with Non-Geostationary satellite system
NGSO	Non-Geostationary Satellite Orbit
OATS	Open Area Test Site
OTA	Over-the-Air
RED	Radio Equipment Directive
SIT	Satellite Interactive Terminal
S-PCN	Satellite Personal Communications Network
SUT	Satellite User Terminal
TC	Technical Committee
TR	Technical Report
UE	User Equipment
VMES	Vehicle Mounted Earth Station
VSAT	Very Small Aperture Terminal
WBES	Fixed and in-motion Wide Band Earth Stations communicating with non-geostationary satellite systems

4 Objectives and Scope of this study

The study contained in the present document represents the next technical phase following ETSI TR 104 305 [i.1], building directly on the technology evolution and use-case assessment carried out. While ETSI TR 104 305 [i.1] established how satellite earth station technologies and deployment scenarios are changing, this further study examines the implications of these developments for the structure and coherence of the existing hENs. Its role is to identify where updates, harmonisation, or potential consolidation may be required to ensure that the standards remain aligned with modern operational conditions. The outcomes of the study contained in the present document provide the technical basis for the drafting activities of the selected hENs and guide the subsequent evolution of the hEN framework under TC SES.

The study contained in the present document acts as the bridge between the technology and use-case evolution identified in ETSI TR 104 305 [i.1] and the concrete revision work to be undertaken in subsequent phases of the study. Its role is to assess how emerging deployment scenarios, modern antenna platforms, and multi-orbit operation affect the structure and coherence of existing hENs, identifying where requirements remain adequate, where transversal updates are needed, and where consolidation of standards may improve clarity and long-term maintainability. By analysing overlaps in architecture, measurement methods, emission behaviour, and operational environments, the study contained in the present document determines coherent groups of hENs for harmonisation or merging and prioritizes those requiring urgent updates. The present document provides the technical foundation for further detailed review of the selected hENs, guiding the scope, structure, and sequencing of the drafting work and enabling the development of revised hENs that are robust, future-proof, and aligned with RED essential requirements.

The present document builds directly upon the analytical foundation established in ETSI TR 104 305 [i.1] and serves as the bridge between the identification of technology-driven update needs and the structured revision of Harmonised Standards (hENs) to be carried out subsequently. While ETSI TR 104 305 [i.1] focuses on expert input collection, technology evolution analysis, and the identification of required updates across the hENs under review, the present document transforms these findings into deployment-driven scenarios, cross-standard mappings, and strategic decisions regarding the potential merging or consolidation of hENs. The present document therefore plays a pivotal role in ensuring that the revision process is coherent and aligned with the operational realities of modern satellite earth station technologies.

The use cases and technology evolution identified in ETSI TR 104 305 [i.1] translates into deployment scenarios across GEO, MEO, and LEO satellite networks, and across maritime, aeronautical, land mobile, fixed, tracking, and narrowband earth station categories. By mapping these deployment scenarios to the hENs under review, the present document identifies where standards overlap, where they diverge, and where consolidation would improve clarity, reduce redundancy, and simplify compliance for manufacturers, notified bodies, and laboratories. This mapping also enables to determine which hENs require update, which require structural reorganization, and which may be merged into unified standards addressing multiple operational environments or frequency bands.

The present document considers the consistency of requirements and measurement methods across the hENs, using the semantic clustering and cross-EN consolidation matrix produced in ETSI TR 104 305 [i.1] as a basis. This evaluation identifies transversal issues-such as antenna-related provisions, off-axis EIRP masks, spurious emission limits, receiver performance requirements, CMF references, and test site definitions-that should be harmonised across standards.

The present document defines the strategic roadmap for hEN revision, based on the prioritization of expert comments and the urgency assessments provided by ETSI TR 104 305 [i.1], establishing a clear sequence for updating the hENs, identifying which standards should be revised first, which require deeper structural changes, and which can be addressed through harmonisation or consolidation. This roadmap guides Experts Group in the drafting of revised hENs and ensures that the revision process is both technically justified and operationally feasible.

The present document transforms the analytical results of ETSI TR 104 305 [i.1] into a coherent strategy for updating, harmonising, and future potentially merging the hENs under the responsibility of ETSI TC SES. It ensures that the revision process reflects real deployment conditions, aligns with modern satellite technologies, and supports long-term regulatory clarity under the Radio Equipment Directive. The analysis of the present document is therefore essential for bridging the gap between expert-driven analysis and the practical drafting of optimized Harmonised Standards (hENs).

The present document therefore encompasses the following main activities: the expansion of ETSI TR 104 305 [i.1] use cases into deployment scenarios; the mapping of these scenarios to the hENs under review; the identification and justification of hEN consolidation opportunities; and the formulation of a structured revision roadmap for subsequent updates of the selected hENs. The present document does not revise the hENs itself, but it provides the analytical and strategic framework that enables further experts standardization activities to produce technically robust, harmonised, and future-proof standards.

5 Methodology for the mapping and selection of priorities hENs

The methodology applied follows a structured analytical process designed to identify transversal alignment needs across the hEN framework and to determine whether any standards exhibit sufficient convergence to warrant further assessment. The approach builds directly on the operational contexts, technological developments, and measurement considerations established in ETSI TR 104 305 [i.1].

Firstly the operational conditions described in the use cases and deployment scenarios are examined. This step allowed identification of functional aspects-such as dynamic pointing, scan-dependent gain, stabilization logic, and NGSO tracking-that appear across multiple hENs and therefore represent transversal impacts.

The analysis then focused on technological evolution, particularly the adoption of electronically steerable antennas and advanced receiver architecture. Then, the analysis continues by reviewing how these technologies are addressed in each hEN and identified cases where comparable functional characteristics are treated with differing clause structures or definitions, indicating the need for aligned architectural and antenna-related provisions.

A third component of the methodology examined measurement practices across the hEN framework. OTA configurations, off-axis assessment methods, and receiver evaluation procedures were compared to determine where legacy approaches no longer reflect modern terminal functionalities and where multiple hENs rely on similar measurement structures with minor variations.

The analysis also conducted a structural comparison of the hENs under revision, analysing clause organization, definitions, transversal elements, and the treatment of emission-related and receiver-related topics. This structural mapping provided the basis for identifying clusters of standards addressing comparable technical content.

The analysis established objective criteria-structural, functional, technological, and measurement-related-to evaluate whether any hENs exhibit a level of convergence that may justify further assessment. These criteria define the analytical framework used without presupposing any specific outcome.

6 Deployment Scenarios

6.0 General

The deployment scenarios translate the use cases and technology evolution identified in ETSI TR 104 305 [i.1] into concrete operational contexts that reflect how satellite earth stations are deployed across GEO, MEO, and LEO networks. These scenarios provide the operational foundation required to understand how the targeted hENs apply in practice, where their scopes overlap, and where inconsistencies or redundancies emerge. By grounding the analysis in real deployment conditions, the analysis ensures that subsequent decisions regarding harmonisation, restructuring, or consolidation of hENs are technically justified and aligned with the realities of modern satellite systems.

6.1 Fixed Earth Stations (GEO)

Fixed earth stations operating in geostationary networks represent the most traditional deployment scenario. These terminals typically rely on mechanically steered antennas, stable pointing geometries, and continuous operation toward a single GEO satellite. Their regulatory constraints, off-axis emission aspects, and measurement methods are well established, but several hENs addressing fixed stations embed assumptions that no longer fully reflect current technology, particularly in relation to flat-array antennas and dynamic payload architectures. The analysis maps these fixed deployments to the existing hENs and identifies where updates are required to accommodate modern antenna technologies and evolving GEO system characteristics.

6.2 Tracking Earth Stations (NGSO)

Tracking earth stations supporting non-geostationary constellations operate with highly dynamic pointing geometries, frequent handovers, Doppler variations, and time-varying interference conditions. The assumptions embedded in several hENs do not fully capture the operational ambient of NGSO tracking terminals, especially when electronically steerable antennas are used. The analysis covers how NGSO tracking deployments interact with the requirements and measurement methods of the relevant hENs and identifies where updates or harmonisation are necessary to ensure regulatory clarity.

6.3 Land mobile earth stations

Land mobile earth stations operate on vehicles or portable platforms and may rely on mechanically steered antennas or electronically steerable arrays. Their mobility introduces rapid changes in orientation, blockage, and multipath conditions, which directly affect off-axis emissions, receiver performance, and control and monitoring functions. The analysis covers evaluation of how land mobile deployments map to the hENs under review and identifies where requirements or measurement methods need to be updated to reflect modern land-based mobility patterns.

6.4 Maritime earth stations

Maritime earth stations installed on vessels experience continuous motion, dynamic roll and pitch, and varying environmental conditions. Historically, maritime earth stations were addressed by separate hENs depending on frequency band, but the analysis also identifies that modern deployments increasingly share common antenna architectures, mobility patterns, and operational constraints across Ku-band and Ka-band systems. The analysis therefore examines how maritime deployments interact with the hENs under review and identifies where consolidation or harmonisation may be justified.

6.5 Aeronautical earth stations

Aeronautical earth stations installed on aircraft operate under highly dynamic conditions, including rapid changes in attitude, high-speed movement, and complex blockage environments. Modern aeronautical terminals increasingly rely on electronically steerable antennas and multi-orbit operation, which challenge several assumptions embedded in existing hENs. The present document also analyses how aeronautical deployments map to the relevant hENs and identifies where updates or structural reorganization are required to ensure consistency with current technologies.

6.6 Narrowband and low-bit-rate terminals

Narrowband and low-bit-rate terminals operating in LEO or GEO networks support specific services such as IoT connectivity, messaging, and low-rate data communications. Their operational characteristics differ significantly from broadband terminals, particularly in terms of waveforms, receiver sensitivity, and interference protection requirements. The present document also evaluates how narrowband deployments interact with the hENs under review and identifies where updates or harmonisation are needed to ensure that requirements remain aligned with modern narrowband architectures.

6.7 Earth Stations in Motion (ESIM)

Earth Stations in Motion represent one of the most technologically dynamic categories addressed in ETSI TR 104 305 [i.1]. ESIM terminals deployed on land, maritime, and aeronautical platforms increasingly share common antenna technologies, multi-orbit capabilities, and dynamic network functions. The separation between ESIM hENs based on operational environment or orbit type has become progressively narrower. The present document examines how ESIM deployments map to the hENs under review and identifies where consolidation into unified ESIM standards may improve clarity, reduce redundancy, and simplify compliance.

7 Mapping deployment scenarios to hENs

7.0 General

The purpose of this clause is to map the deployment scenarios defined in clause 6 to the Harmonised Standards (hENs) under review. While ETSI TR 104 305 [i.1] identified the use cases and the technological evolution affecting the hENs considered, the present document examines how each deployment scenario corresponds to specific standards, where overlaps occur, and where gaps or inconsistencies emerge. This mapping provides the analytical foundation for determining whether certain hENs require updates, structural reorganization, or consolidation. It also highlights cases where multiple hENs address similar operational environments but differ only in frequency band, orbit type, or legacy assumptions, thereby supporting the strategic decisions developed later in the present document.

The mapping process considers the operational characteristics of each deployment scenario, including antenna architecture, mobility patterns, orbit types, waveforms, and regulatory constraints. For each scenario, the present document identifies the hENs that currently apply, evaluates whether their scope and requirements remain appropriate, and determines whether modern deployments challenge the assumptions embedded in the standards. This analysis ensures that the revision roadmap developed in the present document is based in real deployment conditions and aligned with the technological evolution documented in ETSI TR 104 305 [i.1].

7.1 Mapping for Fixed Earth Stations (GEO)

Fixed earth stations operating in geostationary networks correspond primarily to hENs that were developed under stable pointing assumptions and continuous operation toward a single GEO satellite. These stations traditionally rely on mechanically steered antennas, although modern deployments increasingly incorporate flat-array antennas and hybrid architectures. The mapping of this deployment scenario to the hENs under review shows that several standards address fixed GEO terminals either directly or indirectly, depending on their frequency band and service category.

The principal hENs associated with fixed GEO earth stations include those covering VSAT terminals, fixed service earth stations, and GEO-based tracking. These standards generally assume static off-axis emission, well-defined antenna radiation patterns, and measurement methods based on stable geometries. The present document identifies that these assumptions remain valid for traditional mechanically steered antennas but may not fully capture the features of modern electronically steerable arrays, which can introduce dynamic variations even in nominally fixed deployments. As a result, certain requirements and measurement procedures may need to be updated to ensure consistency with current antenna technologies.

The present document also observes that some hENs addressing fixed GEO terminals exhibit structural similarities, particularly in their treatment of off-axis EIRP masks, spurious emissions, receiver performance, and control and monitoring functions. These similarities suggest that further alignment is necessary, especially where standards differ only in frequency band or legacy architectural assumptions. The mapping therefore identifies both the standards currently applicable to fixed GEO deployments and the areas where updates are required to maintain regulatory clarity and technological relevance.

7.2 Mapping for Tracking Earth Stations (NGSO)

Earth stations operating with non-geostationary constellations correspond to a deployment scenario characterized by highly dynamic pointing geometries, continuous satellite motion, and frequent handovers. These terminals should maintain alignment with satellites in LEO or MEO while compensating for Doppler variations, time-varying link conditions, and rapidly changing interference environments. The mapping of this deployment scenario to the hENs under review shows that several standards address NGSO-related functions either directly or indirectly, although many of them were originally drafted under assumptions that reflect earlier generations of NGSO systems.

The hENs associated with NGSO terminals typically include those covering AES operating to NGSO systems, tracking earth stations in Ku-band, and terminals designed for other operations. These standards generally incorporate requirements related to dynamic pointing, off-axis emission control during tracking, and receiver performance under varying link geometries. However, the present document identifies that the operational aspects of modern NGSO systems-particularly those involving electronically steerable antennas-introduce features that are not fully captured in existing hENs. These include instantaneous beam switching, adaptive tracking algorithms, multi-satellite visibility, and rapid transitions between satellites, all of which influence off-axis emission characteristics, measurement procedures, and control and monitoring functions.

The present document also observes that several hENs addressing NGSO terminals share structural similarities with GEO-based standards, especially in their treatment of emission masks, spurious limits, and test site definitions. While these similarities provide a basis for alignment, they also reveal gaps where NGSO-specific operational conditions require distinct provisions. For example, measurement methods designed for static or quasi-static geometries may not adequately represent the dynamic pointing features of NGSO terminals, and certain receiver-related clauses may not reflect the time-varying signal conditions inherent to LEO and MEO systems.

The mapping therefore identifies both the standards currently applicable to NGSO deployments and the areas where updates or harmonisation are required. It highlights the need to incorporate modern antenna technologies, dynamic tracking functions, and multi-orbit operational aspects into the relevant hENs. This analysis supports the broader objective of determining whether NGSO-related standards should be updated individually, aligned with GEO-based standards, or considered for merger into unified multi-orbit hENs.

7.3 Mapping for Land Mobile Earth Stations

Land mobile earth stations correspond to a deployment scenario in which terminals operate on vehicles or portable platforms and are subject to continuous changes in orientation, blockage, and surrounding environmental conditions. These terminals may employ mechanically steered antennas or electronically steerable arrays, and their operational characteristics differ significantly from fixed or maritime deployments due to the variability introduced by land-based mobility. The mapping of this scenario to the hENs under review shows that several standards address land mobile terminals either directly or indirectly, although many of them were originally drafted under assumptions that reflect earlier generations of mobile earth station technologies.

The hENs associated with land mobile earth stations typically include those covering vehicle-mounted terminals, mobile earth stations operating in Ku-band or Ka-band, and certain ESIM standards that encompass land-based mobility. These standards generally incorporate requirements related to off-axis emission control during motion, antenna stabilization functions, and receiver performance under fluctuating link conditions. The present document identifies that modern land mobile deployments introduce operational aspects that are not fully represented in existing hENs, particularly when electronically steerable antennas are used. These antennas may exhibit instantaneous pointing adjustments, dynamic beam shaping, and adaptive control functions that influence emission characteristics and measurement procedures.

Several hENs addressing land mobile terminals share structural similarities with maritime and aeronautical standards, especially in their treatment of emission masks, spurious limits, and control and monitoring functions. While these similarities provide a basis for harmonisation, they also reveal gaps where land-specific operational conditions require distinct provisions. For example, measurement methods designed for stable or predictable motion patterns may not adequately represent the rapid orientation changes and intermittent blockage conditions typical of land-based platforms. Certain receiver-related clauses may also require updates to reflect the signal variations encountered in urban or suburban environments.

The mapping therefore identifies both the hENs currently applicable to land mobile deployments and the areas where updates are required. It highlights the need to incorporate modern antenna technologies, dynamic stabilization functions, and land-specific operational characteristics. This analysis supports the broader objective of determining whether land mobile standards should be updated and aligned with other mobility-related hENs.

7.4 Mapping for Maritime Earth Stations

Maritime earth stations correspond to a deployment scenario in which terminals operate on vessels subject to continuous motion, including roll, pitch, yaw, and varying environmental conditions such as wind, humidity, and sea state. These terminals traditionally rely on mechanically steered antennas equipped with stabilization systems designed to maintain alignment with GEO satellites. However, modern maritime deployments increasingly incorporate electronically steerable antennas and multi-orbit capabilities, which introduce operational characteristics that extend beyond the assumptions embedded in several existing hENs. The mapping of this scenario to the hENs under review shows that several hENs address maritime terminals, often differentiated only by frequency band or legacy architectural considerations.

The hENs associated with maritime earth stations typically include those covering Earth Stations on Vessels operating in Ku-band or Ka-band, as well as certain ESIM standards that encompass maritime mobility. These hENs generally incorporate requirements related to off-axis emission control during vessel motion, antenna stabilization functions, and receiver performance under fluctuating link conditions. Modern maritime deployments introduce features such as instantaneous pointing adjustments, dynamic beam management, and adaptive control functions that are not fully represented in existing hENs, particularly when electronically steerable antennas are used. These aspects influence emission characteristics, measurement procedures, and control and monitoring functions, and therefore require careful consideration during the revision process.

Several hENs addressing maritime terminals share structural similarities with land mobile and aeronautical standards, especially in their treatment of emission masks, spurious limits, and test site definitions. While these similarities provide a basis for alignment, they also reveal gaps where maritime-specific operational conditions require distinct provisions. For example, measurement methods designed for stable or predictable motion patterns may not adequately represent the continuous and multidimensional movement experienced by vessels. Certain receiver-related clauses may also require updates to reflect the signal variations encountered in open-sea environments, including rapid changes in elevation angle and intermittent blockage caused by vessel superstructures.

The mapping therefore identifies both the hENs currently applicable to maritime deployments and the areas where updates are required. It highlights the need to incorporate modern antenna technologies, dynamic stabilization functions, and maritime-specific operational characteristics. This analysis supports the broader objective of determining whether maritime standards should be updated individually or aligned with other mobility-related hENs.

7.5 Mapping for Aeronautical Earth Stations

Aeronautical earth stations correspond to a deployment scenario in which terminals operate on aircraft subject to highly dynamic motion, rapid changes in attitude, and complex blockage environments. The terminals should maintain stable communication links despite variations in pitch, roll, yaw, and flight trajectory, and they often rely on advanced stabilization systems or electronically steerable antennas to compensate for these conditions. The mapping of this scenario to the hENs under review shows that several hENs address aeronautical terminals, although many were originally drafted under assumptions that reflect earlier generations of airborne earth station technologies.

The hENs associated with aeronautical earth stations typically include those covering AES operating in Ku-band or Ka-band, as well as ESIM standards that encompass aeronautical mobility. These standards generally incorporate requirements related to off-axis emission control during aircraft motion, antenna stabilization functions, and receiver performance under rapidly changing link geometries. Modern aeronautical deployments introduce operational characteristics that are not fully represented in existing hENs, particularly when electronically steerable antennas are used. These antennas may exhibit instantaneous pointing adjustments, dynamic beam management, and adaptive control functions that influence emission aspects, measurement procedures, and control and monitoring functions.

Several hENs addressing aeronautical terminals share structural similarities with maritime and land mobile standards, especially in their treatment of emission masks, spurious limits, and test site definitions. While these similarities provide a basis for harmonisation, they also reveal gaps where aeronautical-specific operational conditions require distinct provisions. For example, measurement methods designed for stable or predictable motion patterns may not adequately represent the rapid attitude changes and intermittent blockage conditions typical of aircraft operation. Certain receiver-related clauses may also require updates to reflect the signal variations encountered at high altitude, including changes in elevation angle, atmospheric attenuation, and transitions between coverage beams.

The mapping therefore identifies both the hENs currently applicable to aeronautical deployments and the areas where updates are required. It highlights the need to incorporate modern antenna technologies, dynamic stabilization functions, and aeronautical-specific operational characteristics into the relevant hENs. This analysis supports the broader objective of determining whether aeronautical hENs should be updated individually or aligned with other mobility-related hENs.

7.6 Mapping for Narrowband and Low Bit Rate Terminals

Narrowband and low bit rate terminals correspond to a deployment scenario in which earth stations support services such as IoT connectivity, messaging, telemetry, and low-rate data communications. These terminals typically operate in LEO or GEO networks and rely on waveforms and link configurations that differ substantially from those used in broadband systems. Their operational characteristics include reduced bandwidth, specific modulation formats, stringent receiver sensitivity requirements, and distinct interference protection needs. The mapping of this scenario to the hENs under review shows that only a limited number of hENs explicitly address narrowband terminals, and several of them were developed under assumptions that reflect earlier generations of low-rate satellite services.

The hENs associated with narrowband and low bit rate terminals generally include those covering Mobile Satellite Service terminals operating below 1 GHz and certain hENs addressing narrowband uplink and downlink configurations in higher frequency bands. These hENs incorporate requirements related to receiver sensitivity, adjacent band selectivity, and interference protection mechanisms tailored to low-rate services. Modern narrowband deployments introduce operational aspects that are not fully represented in existing hENs, particularly in relation to updated waveforms or adaptive coding schemes. These features influence receiver performance, interference protection clauses, and measurement procedures, and therefore require careful consideration during the revision process.

Narrowband hENs differ structurally from broadband standards, especially in their treatment of emission limits, test site definitions, and control and monitoring functions. While these differences reflect the distinct nature of narrowband services, they also reveal gaps where certain provisions may need to be aligned with broadband hENs to ensure consistency across the overall standardization framework. For example, measurement methods designed for legacy narrowband waveforms may not adequately represent the characteristics of modern IoT-oriented signals, and certain receiver-related clauses may require updates to reflect the operational conditions encountered in dense LEO constellations or hybrid GEO-LEO architectures.

The mapping therefore identifies both the hENs currently applicable to narrowband and low bit rate deployments and the areas where updates are required. It highlights the need to incorporate modern waveform characteristics, updated receiver performance criteria, and narrowband-specific operational conditions. This analysis supports the broader objective of determining whether narrowband hENs should be updated individually or aligned with broadband hENs where appropriate.

7.7 Mapping for Earth Stations in Motion (ESIM)

Earth Stations in Motion represent one of the most technologically dynamic deployment scenarios addressed in the present document. ESIM terminals operate on land, maritime, and aeronautical platforms and should maintain stable communication links while subject to continuous motion, rapid orientation changes, and varying blockage conditions. Modern ESIM systems increasingly rely on electronically steerable antennas, multi-orbit capabilities, adaptive tracking functions, and advanced control mechanisms that allow seamless transitions between GEO, MEO, and LEO satellites. The mapping of this scenario to the hENs under review shows that several hENs address ESIM terminals, although many were developed when ESIM technologies were less mature and when multi-orbit operation was not yet widespread.

The hENs associated with ESIM terminals typically include those covering land ESIM, maritime ESIM, aeronautical ESIM, and certain NGSO-specific hENs that encompass mobile operation. These hENs incorporate requirements related to off-axis emission control during motion, antenna stabilization functions, and receiver performance under dynamic link geometries. Modern ESIM deployments introduce operational characteristics that extend beyond the assumptions embedded in existing hENs. Electronically steerable antennas may exhibit instantaneous pointing adjustments, dynamic beam shaping, and adaptive control functions that influence emission aspects, measurement procedures, and control and monitoring functions. Multi-orbit operation further introduces transitions between satellites with different elevation angles, link budgets, and interference environments, which are not fully represented in several legacy hENs.

ESIM-related hENs share structural similarities with maritime, aeronautical, and land mobile standards, particularly in their treatment of emission masks, spurious limits, and test site definitions. While these similarities provide a basis for harmonisation, they also reveal gaps where ESIM-specific operational conditions require distinct provisions. For example, measurement methods designed for single-orbit or quasi-static geometries may not adequately represent the dynamic pointing features and rapid satellite transitions typical of multi-orbit ESIM systems. Certain receiver-related clauses may also require updates to reflect the signal variations encountered during transitions between satellites, including changes in Doppler characteristics, link availability, and interference protection requirements.

The mapping therefore identifies both the standards currently applicable to ESIM deployments and the areas where updates are required. It highlights the need to incorporate modern antenna technologies, multi-orbit operational functions, and ESIM-specific dynamic characteristics into the relevant hENs. This analysis supports the broader objective of determining whether ESIM hENs should be updated individually or aligned across mobility categories.

8 Alignment and merging assessment across hENs

8.0 General

The analysis performed in ETSI TR 104 305 [i.1] has revealed a set of transversal technological and measurement-related evolutions that affect multiple hENs simultaneously. These evolutions-covering antenna architectures, dynamic operating conditions, emission aspects, receiver performance, and updated test methodologies-create the need for a coordinated alignment of requirements, definitions, and measurement procedures across several standards. The present document builds on this foundation by identifying where such alignment is necessary, mapping the relevant use cases and deployment scenarios to the affected hENs, and determining the structural areas where updates should be applied consistently to maintain clarity and long-term applicability.

The present document also evaluates whether the technical convergence identified in ETSI TR 104 305 [i.1] justifies a potential future merging of hENs. This assessment is deliberately limited in scope and only considered when the overlap in technologies, deployment conditions, and clause structures is sufficiently strong to support a unified deliverable. Together, the alignment activities and the targeted updates assessment provide a clear, structured basis for updating the hENs in a coherent and future-proof manner.

8.1 Drivers for alignment and structural review

8.1.0 General

The need for transversal alignment across several hENs arises directly from the technological and methodological evolution identified in ETSI TR 104 305 [i.1]. Modern antenna platforms (particularly electronically steerable arrays) exhibit operating characteristics that are increasingly similar across different categories of earth stations. These include comparable steering dynamics, pattern evolution, control functions, and monitoring capabilities, regardless of whether the terminals operate in fixed, maritime, aeronautical, land-mobile, or multi-orbit environments. As these technologies converge, the corresponding requirements, definitions, and measurement provisions in multiple hENs should be aligned to ensure consistent interpretation and long-term applicability.

A second driver relates to the evolution of measurement methodologies. ETSI TR 104 305 [i.1] highlights that several test procedures, test site configurations, and measurement setups used across different hENs are based on legacy assumptions that no longer reflect modern terminal architectures or dynamic operating conditions. Updated over-the-air methods, revised blocking and selectivity evaluations, and improved approaches for off-axis emission assessment are required across multiple hENs. Aligning these measurement provisions is essential to avoid divergence in test practices and to ensure that similar technologies are evaluated using coherent and technically robust procedures.

Deployment scenarios also contribute to the need for alignment. ETSI TR 104 305 [i.1] shows that earth stations operating in motion-whether aeronautical, maritime, land-mobile, or NGSO-tracking-share functional aspects such as stabilization logic, dynamic pointing, and real-time adaptation to link conditions. These commonalities indicate that several hENs should incorporate aligned definitions, updated operational assumptions, and consistent interpretations of dynamic conditions.

Finally, the structural analysis performed in ETSI TR 104 305 [i.1] (macro-clusters and sub-clusters) reveals that many hENs share similar clause structures, emission frameworks, and receiver-related provisions. This cross-standard consistency reinforces the need for coordinated alignment and also informs the identification of cases where technical convergence may justify a merging of selected hENs, although such merging remains a secondary activity.

8.1.1 Technology evolution drivers

ETSI TR 104 305 [i.1] identifies several technological evolutions that directly motivate the need for transversal alignment across multiple hENs. Modern antenna platforms introduce operating characteristics that differ substantially from the assumptions embedded in several existing standards. These platforms rely on dynamic steering mechanisms, adaptive radiation patterns, multibeam operation, and real-time optimization of pointing and link conditions. As these functional aspects appear across a wide range of earth station categories, the corresponding requirements and definitions should be aligned to ensure consistent treatment of comparable technologies.

The transition toward multi-orbit operation further reinforces this need. Terminals capable of operating with GEO, MEO, and LEO systems exhibit similar dynamic conditions, including continuous tracking, rapid pointing adjustments, and frequent changes in link geometry. ETSI TR 104 305 [i.1] shows that these elements are present in several use cases and deployment scenarios, indicating that multiple hENs should incorporate aligned provisions addressing dynamic pointing, stabilization logic, and operational constraints associated with multi-orbit connectivity.

ETSI TR 104 305 [i.1] also highlights the evolution of waveform processing, modem architecture, and receiver performance. Advanced coding schemes, adaptive modulation, improved interference mitigation techniques, and enhanced receiver sensitivity appear across different categories of earth stations. These developments require aligned updates to receiver-related clauses, adjacent-band selectivity provisions, and blocking performance requirements to ensure that similar technological capabilities are addressed coherently across the affected hENs.

Finally, the increasing use of flat-array antennas introduces specific functional characteristics-such as pattern shaping, scan-dependent gain variation, and electronically controlled polarization-that are relevant to several standards. ETSI TR 104 305 [i.1] demonstrates that these aspects should be reflected through aligned antenna-related provisions, updated definitions, and consistent structural treatment across the hEN framework.

8.1.2 Measurement methodology drivers

ETSI TR 104 305 [i.1] identifies several methodological evolutions that directly motivate the need for transversal alignment across multiple hENs. A significant number of existing measurement procedures were developed under assumptions linked to earlier generations of earth stations, where antenna architectures, dynamic operating conditions, and receiver capabilities were substantially different from those used today. As modern terminals incorporate electronically steerable arrays, multi-orbit operation, and advanced signal processing, the corresponding test methods should be updated and aligned across the affected standards.

One of the main drivers is the evolution of over-the-air measurement techniques. ETSI TR 104 305 [i.1] shows that several hENs rely on test site configurations and procedures that do not fully capture the operating characteristics of flat-array antennas or dynamically steered terminals. Updated OATS methodologies-covering scan-dependent radiation patterns, adaptive pointing, and real-time control functions-are required across multiple standards, and their alignment is essential to ensure consistent evaluation of comparable technologies.

A second driver concerns emission-related measurement procedures, including off-axis EIRP assessment, spurious emissions, and adjacent-band selectivity. ETSI TR 104 305 [i.1] highlights that these procedures are implemented with similar structures across several hENs, but with differences in definitions, numerical parameters, and test setups. Aligning these provisions will improve coherence and reduce discrepancies in how emission limits are verified for terminals with comparable functional characteristics.

Receiver-related evaluations also require alignment. ETSI TR 104 305 [i.1] identifies that blocking performance, sensitivity thresholds, and interference-mitigation capabilities have evolved significantly, particularly for terminals operating in dynamic or multi-orbit conditions. Several hENs address these aspects using legacy formulations that do not reflect modern receiver architectures. Updated and aligned measurement procedures are needed to ensure that receiver performance is assessed consistently across standards.

ETSI TR 104 305 [i.1] shows that test site descriptions, measurement setups, and cross-referencing practices vary across the hENs under review, despite addressing similar technologies. Aligning these transversal elements - such as definitions of test environments, calibration requirements, and references to external measurement standards - will support clarity, reduce redundancy, and facilitate the drafting work in Experts Group.

8.1.3 Deployment scenario drivers

Several deployment scenarios share operational characteristics that directly motivate transversal alignment across multiple hENs. Earth stations operating in motion whether aeronautical, maritime, land-mobile, or vehicle-mounted rely on stabilization mechanisms, dynamic pointing, and continuous adaptation to link conditions. These functional aspects appear repeatedly across the use cases analysed in ETSI TR 104 305 [i.1], indicating that some hENs should incorporate aligned provisions addressing dynamic operating conditions, pointing constraints, and real-time control functions.

Terminals supporting NGSO connectivity introduce additional drivers. The LEO and MEO operation require rapid tracking, frequent changes in link geometry, and continuous updates of antenna orientation. These elements are present in multiple hENs, often treated with similar clause structures but differing assumptions or definitions. Aligning these provisions is necessary to ensure consistent interpretation of dynamic conditions across standards addressing comparable deployment environments.

The fixed and transportable earth stations increasingly share functional aspects with mobile platforms, particularly when using electronically steerable antennas. These similarities indicate that hENs under review should adopt aligned definitions of operational states, updated descriptions of deployment conditions, and consistent treatment of dynamic or semi-dynamic operation. Together, these drivers demonstrate that deployment scenarios are a key source of transversal alignment needs across the hEN framework.

8.1.4 Cross-EN structural drivers (from ETSI TR 104 305 clusters)

The structural analysis of hEN under review performed in ETSI TR 104 305 [i.1]-based on macro-clusters, technical sub-clusters, and EN-by-EN cross-mapping-reveals a high degree of similarity. Many hENs share comparable clause structures, similar approaches to emission limits, aligned receiver-related provisions, and analogous descriptions of control and monitoring functions. These structural consistencies indicate that transversal alignment is both technically justified and necessary to maintain coherence across the hENs.

ETSI TR 104 305 [i.1] identifies clusters where multiple hENs address the same technical topics using slightly different formulations, numerical parameters, or definitions. Examples include off-axis EIRP limits, spurious emission clauses, adjacent-band selectivity, antenna pattern descriptions, and OATS measurement setups. Aligning these transversal elements will reduce inconsistencies, improve clarity, and support the review of the hENs.

The clustering analysis also highlights cases where the structural overlap between hENs is sufficiently strong to justify evaluating a potential merging. This includes similarities in clause organization, functional assumptions, and measurement provisions. These structural drivers provide the technical basis for identifying the hENs to be assessed for potential merging.

8.2 Identification of transversal alignment activities

8.2.0 General

ETSI TR 104 305 [i.1] demonstrates that several technological, operational, and measurement-related evolutions affect multiple hENs in a consistent manner. These cross-standard impacts require a set of transversal alignment activities to ensure that similar technologies, operating conditions, and test methodologies are addressed coherently across the relevant deliverables. The purpose of this clause is to identify those activities and define the areas where alignment is technically justified and necessary.

A first group of alignment activities concerns architectural assumptions and functional aspects common to modern earth stations. Electronically steerable antennas, multi-orbit operation, dynamic pointing mechanisms, and real-time adaptation to link conditions appear across several use cases and deployment scenarios. These elements should be reflected through aligned definitions, updated operational assumptions, and consistent clause structures in the affected hENs.

A second group relates to measurement methods and test setups. ETSI TR 104 305 [i.1] highlights that hENs under review rely on legacy procedures that do not fully capture the operating characteristics of flat-array antennas or dynamically steered terminals. Updated over-the-air techniques, revised approaches for off-axis emission assessment, improved receiver-related evaluations, and clearer descriptions of test site configurations are required across multiple standards. Aligning these provisions ensures that comparable technologies are evaluated using coherent and technically robust methodologies.

A third group of alignment activities concerns antenna-related provisions, including transmit and receive radiation patterns, scan-dependent gain variation, dynamic pointing conditions, and the interpretation of flat-array characteristics. ETSI TR 104 305 [i.1] identifies that several hENs address these aspects using similar but not fully aligned formulations. Updating these clauses in a coordinated manner will improve clarity and reduce inconsistencies across standards.

A fourth group addresses emission-related clauses, particularly off-axis EIRP limits, spurious emissions, and adjacent-band selectivity. ETSI TR 104 305 [i.1] shows that these topics appear repeatedly across the hENs under review, often with comparable structures but differing numerical values, definitions, or measurement interpretations. Alignment is required to ensure consistent treatment of similar emission conditions.

ETSI TR 104 305 [i.1] also identifies the need to align control and monitoring functions, terminology, and definitions across several hENs. These transversal elements support coherent interpretation of requirements and facilitate the further update of the hENs. The structural analysis performed in ETSI TR 104 305 [i.1] confirms that many hENs share comparable clause organization and address identical technical topics with minor variations. Aligning these transversal elements will reduce inconsistencies, improve clarity, and ensure that hENs can be updated in a coherent and future-proof manner.

8.2.1 Alignment of architectural assumptions

The evolution of earth station architectures across multiple use cases has led to a common set of functional characteristics that several hENs should address coherently. Electronically steerable antennas, dynamic pointing mechanisms, and real-time adaptation to link conditions now appear in a wide range of terminal categories. These shared operational features indicate that aligned definitions, updated architectural assumptions, and consistent clause structures are required to ensure that comparable technologies are treated uniformly across the hEN framework.

8.2.2 Alignment of measurement methods and test setups

Recent developments in measurement practices reveal that several existing test procedures and setups no longer reflect the operating conditions of modern terminals. Open Area Test Sites and in general over-the-air techniques, off-axis EIRP assessment methods, receiver-related evaluations, and test site configurations require updates to capture the functional characteristics of electronically steerable antennas and dynamically operated terminals. Establishing aligned measurement provisions across the affected hENs will ensure coherent evaluation of similar technologies and support the further drafting activities

8.2.3 Alignment of antenna-related provisions

Antenna-related requirements appear across numerous hENs with similar intent but differing formulations. Transmit and receive radiation patterns, scan-dependent gain variation, dynamic pointing conditions, and the interpretation of flat-array characteristics are addressed in multiple standards, often with minor structural or definitional differences. Given the widespread adoption of electronically steerable antennas in fixed, mobile, maritime, aeronautical, and NGSO-tracking terminals, aligned antenna-related clauses are needed to ensure clarity and consistency across the hEN framework.

8.2.4 Alignment of emission-related clauses

Emission-related topics-such as off-axis EIRP limits, spurious emissions, and adjacent-band selectivity-are treated across several hENs using comparable clause structures but with variations in numerical parameters, definitions, or measurement interpretations. These differences can lead to inconsistencies when evaluating terminals with similar functional characteristics. Aligning emission-related provisions will provide a coherent basis for interpreting and verifying emission limits across standards addressing comparable technologies and deployment scenarios.

8.2.5 Alignment of Control and Monitoring Functions (CMFs)

Control and Monitoring Functions (CMFs) are addressed in multiple hENs, often with analogous descriptions but differing terminology or structural placement. Modern terminals, especially those operating under dynamic or multi-orbit conditions, depend on real-time control logic, adaptive tracking, and continuous monitoring of link conditions. Aligning CMF-related clauses across the affected standards will support consistent interpretation of these functional aspects and reinforce the structural coherence identified in ETSI TR 104 305 [i.1]'s clustering analysis.

8.2.6 Alignment of terminology and definitions

Terminology and definitions used across the hENs frequently refer to similar technical concepts but with minor variations that can affect clarity and interpretation. Terms related to antenna operation, dynamic conditions, emission limits, receiver performance, and measurement setups appear in multiple standards and should be aligned to ensure consistency. A coordinated update of terminology and definitions will strengthen transversal alignment and facilitate the drafting work, particularly in areas where structural overlap between hENs is significant.

8.3 Mapping of use cases and deployment scenarios to affected hENs

8.3.0 General

The transversal evolution of technologies, operating conditions, and measurement methodologies requires a structured mapping between the use cases and deployment scenarios defined in ETSI TR 104 305 [i.1] and the hENs affected by these developments. This mapping identifies where alignment activities should be applied and clarifies the technical basis for updating the relevant standards. By linking operational contexts, functional characteristics, and measurement requirements to specific hENs, this clause provides the foundation for coherent drafting of the hENs and supports the identification of cases where structural overlap may justify evaluating a potential merging.

The mapping process considers three elements:

1) the operational characteristics associated with each use case;

2) the functional conditions defined in each deployment scenario;

3) the clause structures and technical topics addressed in the corresponding hENs.

This approach ensures that alignment activities are applied consistently across hENs addressing comparable technologies and operational environments.

8.3.1 Mapping of use cases

The use cases describe how earth stations operate under different functional conditions, including fixed operation, mobility, and dynamic pointing. Each use case introduces specific operational characteristics, such as stabilization logic, scan-dependent gain variation, or continuous tracking, that appear across multiple hENs. Mapping these characteristics to the affected standards allows identification of the clauses that require aligned definitions, updated operational assumptions, or revised measurement provisions. This mapping also highlights cases where several hENs address identical functional aspects with minor structural differences, reinforcing the need for transversal alignment.

8.3.2 Mapping of deployment scenarios

Deployment scenarios provide a detailed description of the environments in which earth stations operate, including aeronautical, maritime, land-mobile, transportable, and NGSO-tracking conditions. Many of these scenarios share functional elements such as dynamic pointing, real-time adaptation to link geometry, and continuous monitoring of operating conditions. Mapping these shared elements to the relevant hENs identifies where aligned provisions are required to ensure consistent treatment of comparable deployment environments. This mapping also supports the identification of structural overlaps between standards addressing similar operational contexts.

8.3.3 Mapping of technology evolution

Technological evolution, particularly the adoption of electronically steerable antennas, multibeam operation, and advanced receiver architectures, affects several hENs simultaneously. Mapping these developments to the standards that address similar antenna platforms, receiver capabilities, or control functions allows identification of the clauses that require aligned updates. This mapping also clarifies where technological convergence across hENs is sufficiently strong to justify evaluating a potential merging, based on similarities in clause organization, functional assumptions, and measurement provisions.

8.3.4 Mapping of measurement evolution

The evolution of measurement methodologies introduces transversal impacts across multiple hENs. Updated over-the-air techniques, revised off-axis EIRP assessment methods, improved receiver-related evaluations, and updated test site configurations are relevant to several standards. Mapping these methodological developments to the affected hENs identifies where aligned measurement provisions are required to ensure coherent evaluation of comparable technologies. This mapping also highlights cases where multiple hENs rely on similar measurement structures with minor variations, reinforcing the need for coordinated alignment.

8.3.5 Identification of hENs requiring alignment

The consolidated mapping of use cases, deployment scenarios, technological evolution, and measurement evolution shows that all hENs under review exhibit transversal elements that require coordinated alignment. Although the degree and nature of alignment differ across hENs, each hEN is affected by at least one of the following cross-cutting drivers:

- modern antenna architectures (electronically steerable arrays, scan-dependent gain, dynamic pointing);
- GSO or NGSO operation;
- updated emission-related provisions;
- evolving receiver performance requirements;
- modernized measurement methodologies;
- structural similarities identified through clustering analysis.

To provide clarity and support for future drafting, the hENs requiring alignment are grouped below according to the dominant transversal drivers affecting each standard.

(Group 1) hENs impacted by dynamic operation, electronically steerable antennas, and orbit connectivity

These standards address earth stations whose functional characteristics include stabilization logic, continuous tracking, scan-dependent gain variation, and real-time adaptation to link geometry. Alignment is required to ensure consistent treatment of dynamic operating conditions and antenna-related provisions:

- **ETSI EN 303 979** [i.4] - ESOMP NGSO Ka
- **ETSI EN 303 978** [i.3] - ESOMP GEO Ka
- **ETSI EN 303 984** [i.15] - ESOA NGSO Ku APP30B
- **ETSI EN 302 448** [i.13] - EST (Trains) GEO Ku
- **ETSI EN 302 977** [i.16] -VMES GEO Ku
- **ETSI EN 303 980** [i.5] - NEST (Fixed and in-motion NGSO) Ku

These hENs form a clear transversal cluster in ETSI TR 104 305 [i.1], sharing similar clause structures for dynamic pointing, antenna control functions, and off-axis emission constraints.

(Group 2) hENs impacted by updated emission-related provisions (off-axis EIRP, spurious, adjacent-band selectivity)

These standards exhibit comparable emission frameworks but differ in numerical values, definitions, or measurement interpretations. Alignment is required to ensure coherent verification of emission limits across similar technologies:

- **ETSI EN 303 979** [i.4] - ESOMP NGSO Ka
- **ETSI EN 303 978** [i.3] - ESOMP GEO Ka
- **ETSI EN 303 699** [i.2] - Fixed NGSO Ka
- **ETSI EN 301 459** [i.10] - SIT/SUT GEO Ka

- **ETSI EN 302 340** [i.12] - Maritime ESV Ku
- **ETSI EN 301 447** [i.9] - Maritime ESV GEO (C-band)
- **ETSI EN 301 443** [i.8] - VSAT GEO (C)

These hENs form the strongest emission-related cluster identified in ETSI TR 104 305 [i.1].

(Group 3) hENs impacted by updated receiver-related provisions (blocking, sensitivity, adjacent-band selectivity)

Modern receiver architectures introduce transversal impacts across several standards. Alignment is required to ensure consistent evaluation of receiver performance:

- **ETSI EN 301 443** [i.8] - VSAT GEO C
- **ETSI EN 301 459** [i.10] - SIT/SUT GEO Ka
- **ETSI EN 303 699** [i.2] - Fixed NGSO Ka
- **ETSI EN 303 980** [i.5] - NEST (Fixed and in motion NGSO) Ku

These hENs share similar receiver-related clause structures, as confirmed by ETSI TR 104 305 [i.1] clustering.

(Group 4) hENs impacted by updated measurement methodologies (OTA, off-axis assessment, test site configurations)

These hENs rely on legacy measurement setups that require alignment to reflect modern antenna platforms and dynamic operation:

- **ETSI EN 303 979** [i.4] - ESOMP NGSO Ka
- **ETSI EN 303 978** [i.3] - ESOMP GEO Ka
- **ETSI EN 303 984** [i.15] - ESOA NGSO Ku APP30B
- **ETSI EN 302 977** [i.16] - VMES GEO Ku
- **ETSI EN 302 340** [i.12] - Maritime ESV Ku
- **ETSI EN 301 447** [i.9] - Maritime ESV GEO (C band)

These hENs form the OTA-related cluster identified in ETSI TR 104 305 [i.1].

(Group 5) hENs impacted by terminology, definitions, and clause-structure alignment

These standards share similar clause organization, definitions, and transversal elements such as control and monitoring functions. Alignment is required to maintain structural coherence across the hEN framework:

- **ETSI EN 301 427** [i.6] - Low-data-rate MES
- **ETSI EN 302 574-3** [i.14] - Narrowband MES
- **ETSI EN 301 442** [i.7] - NGSO handheld MES
- **ETSI EN 301 721** [i.11] - LBRDC MES

These hENs form the "narrowband and handheld MES" cluster in ETSI TR 104 305 [i.1].

The hENs under review require alignment, though the nature and intensity of alignment differ across clusters. This mapping:

- identifies the transversal drivers affecting each hEN;
- clarifies the structural basis for alignment;

- and highlights the subset of hENs where structural overlap is sufficiently strong to justify evaluating a potential merging.

8.4 Assessment of potential merger of selected hENs

8.4.0 General

The present document has evaluated whether any of the hENs under revision exhibit sufficient structural, functional, technological, or measurement-related convergence to justify assessing a potential merging which could be considered by TC SES as a subsequent activity further to the present document. The objective of this clause is to apply the criteria defined in clause 8.4.1 to several groups of hENs that present partial similarities, and to determine whether these similarities are strong enough to support consolidation without compromising clarity, applicability, or regulatory coherence.

8.4.1 Criteria for evaluating technical convergence

The assessment applies four criteria derived from the transversal analysis in clauses 8.1 and 8.2: structural alignment, functional alignment, technological alignment, and measurement alignment. Only hENs meeting all four criteria to a sufficient degree can be considered candidates for merging.

8.4.2 Assessment case 1: ETSI EN 303 979 and ETSI EN 303 699

ETSI EN 303 979 [i.4] and ETSI EN 303 699 [i.2] both address NGSO Ka-band earth stations and rely on electronically steerable antennas with comparable dynamic pointing behaviour. Their clause structures, emission frameworks, receiver-related provisions, and measurement methodologies show a high degree of similarity. The analysis indicates that these two hENs exhibit strong convergence across all four criteria and therefore constitute a technically viable case for evaluating potential merging.

8.4.3 Assessment case 2: ETSI EN 301 443 and ETSI EN 301 447

ETSI EN 301 443 [i.8] and ETSI EN 301 447 [i.9] operate in C-band and share certain emission-related structures. However, their deployment scenarios differ substantially: ETSI EN 301 443 [i.8] addresses VSAT terminals, while ETSI EN 301 447 [i.9] covers maritime earth stations on vessels. These differences affect antenna control functions, installation constraints, environmental conditions, and measurement setups. Although partial structural and technological similarities exist, the functional and operational divergence is significant. The level of convergence is therefore insufficient to support merging.

8.4.4 Assessment case 3: ETSI EN 301 427, ETSI EN 302 340, ETSI EN 302 448 and ETSI EN 302 977

These four hENs operate in Ku-band and address earth stations with mobility or tracking capabilities. They share transversal elements such as stabilization logic, dynamic pointing, off-axis control, and antenna behaviour under motion. Although each hEN is tailored to a distinct platform category (low-data-rate MES, maritime earth stations on vessels, train-mounted earth stations, and vehicle-mounted earth stations) the underlying functional principles are comparable.

The analysis shows that the differences between platforms are primarily related to installation constraints, environmental conditions, and operational profiles, rather than to the fundamental behaviour of the antenna system or the emission and measurement frameworks. As a result, these differences could be treated as platform-specific operational profiles within a single consolidated hEN, provided that the document adopts a modular structure with a common technical core and dedicated sections for each platform type.

A potential merging would therefore be technically feasible under certain conditions. These include defining a unified concept of "earth station with mobility", consolidating transversal antenna-related and measurement provisions, maintaining platform-specific installation and environmental requirements as separate profiles, and validating numerical parameters to ensure that consolidation does not broaden or narrow applicability. If these conditions are met, the four hENs could be integrated into a single document without compromising clarity or regulatory precision.

8.4.5 Comparative outcome of the three assessments

The evaluation of the three cases shows that only ETSI EN 303 979 [i.4] and ETSI EN 303 699 [i.2] meet the convergence criteria to a degree that justifies considering merging. The other two cases - ETSI EN 301 443 [i.8] with ETSI EN 301 447 [i.9], and the group formed by ETSI EN 301 427 [i.6], ETSI EN 302 340 [i.12], ETSI EN 302 448 [i.13] and ETSI EN 302 977 [i.16] - present partial similarities but do not achieve the same level of structural, functional, technological, and measurement alignment as compared with the first case.

8.4.6 Structural implications of a potential merging

If merging is pursued for any hEN pair meeting the convergence criteria, the structural implications should be assessed carefully. Merging would require integrating duplicated clauses, unifying definitions, and ensuring consistent treatment of transversal elements identified in clauses 8.1 and 8.2. In the specific case of ETSI EN 303 979 [i.4] and ETSI EN 303 699 [i.2], the high degree of parallelism between the two documents means that structural integration would be limited and manageable. The merged document would retain the same technical scope and reduce redundancy without altering the underlying regulatory intent.

8.4.7 Boundary conditions and limitations applicable to any merging scenario

Any merging scenario should respect several boundary conditions to ensure that merging does not introduce ambiguity or reduce clarity. Distinctions between fixed and in-motion operation should be preserved where relevant, as these differences affect antenna control functions, stabilization logic, and measurement setups. Compatibility with deployment-specific assumptions referenced in the original standards should be maintained, particularly regarding installation constraints and operational environments. Numerical parameters and test configurations should be validated to ensure that consolidation does not unintentionally broaden or narrow applicability. These limitations ensure that any merging remains technically sound and preserves the precision required for earth station regulation.

8.4.8 Implications for subsequent work

8.4.8.0 General

The transversal alignment activities, together with the assessment of potential merging, define the scope and structure of the drafting work to be further undertaken. This clause explains how the aligned provisions, the merging-related conclusions, and the mapping performed in clause 8.3 should be integrated into the updated hEN framework. It also outlines the sequence of drafting tasks and the coordination required to ensure coherent implementation of the present document recommendations and outcomes.

8.4.8.1 Consolidation of aligned provisions

The further update of hENs will need to incorporate the aligned architectural assumptions, antenna-related provisions, emission clauses, receiver-related requirements, and measurement methodologies identified in clause 8.2. This consolidation requires integrating consistent definitions, updating clause structures to reflect unified operational assumptions, and ensuring coherent treatment of dynamic pointing, scan-dependent gain, NGSO tracking, and other transversal elements. Experts Group should also reflect the updated measurement setups, including revised OTA configurations, off-axis assessment logic, and receiver evaluation methods. The result will be a coherent set of hENs with reduced redundancy and improved clarity across the framework.

8.4.8.2 Integration of merging-related outcomes

The merging assessment presented in clause 8.4 indicates that ETSI EN 303 979 [i.4] and ETSI EN 303 699 [i.2] exhibit sufficient convergence to justify evaluating consolidation into a single hEN. Experts Group should therefore prepare a unified clause structure covering NGSO Ka-band earth stations, integrate duplicated provisions into a single consolidated set of requirements, and maintain distinctions between fixed and in-motion operation where technically necessary. Numerical parameters and measurement configurations should be validated to ensure that consolidation does not broaden or narrow applicability. This integration will streamline the regulatory landscape for NGSO Ka-band terminals and simplify future maintenance.

8.4.8.3 Drafting sequence and document dependencies

The drafting should follow a structured sequence to ensure coherence across all affected hENs. Transversal alignment should be applied first, ensuring that all integrated provisions identified in clause 8.3.5 are reflected consistently. Measurement annexes should then be updated to reflect unified setups, off-axis assessment methods, and receiver-related evaluations. Definitions and terminology should be revised to ensure consistency across the entire resulting hEN, followed by cross-document validation to confirm that aligned provisions do not introduce contradictions or gaps.

8.4.9 Other implications of potential merger of some hENs

It has been indicated in previous clauses that merger of several existing hENs would be possible or even recommendable from a strictly technical viewpoint. Considerations of the technical reasons are provided in the present document and supported by considerations of ETSI TR 104 305 [i.1]. However, the merger of existing hENs should not only consider the technical impacts or benefits, but also the possible implications in the legal and regulatory context where the hENs are referred to, mandated, or simply considered as reference harmonised standard.

Due to the long existence of many hENs, their use has been considered by national regulatory authorities in the EU (and European Economy Space) and around the world. Consequently, the hENs are referred to in the national regulation. Furthermore, satellite operators who obtain licenses and authorizations to operate in each country, use to obtain such licenses and authorization to respect technical parameters of a given relevant hEN. Therefore, the merger of hENs with creation of new hENs or embedding some hENs into the number of other hENs will undoubtedly require actions by interested parties (manufacturers, operators, service providers) to update those records and refer to the updated resulting hENs. This process cannot be considered as an insurmountable barrier but certainly, a cumbersome task which may be more relevant or include higher burden for some hENs than for others.

Therefore, the present document just points towards potential merger of some hENs which TC SES may take into account and consider facing such tastes in the future, while for the short-term update of the targeted hEN, no recommendation to merger hENs is provided.

9 Conclusions

The analysis performed in the present document demonstrates that the evolution of technologies, operating conditions, and measurement methodologies across the TC SES domain introduces transversal impacts affecting a significant portion of the hEN framework of the harmonised earth stations. The mapping of use cases, deployment scenarios, technological developments, and measurement evolution confirms that several hENs address comparable functional aspects, often with similar clause structures and overlapping technical content. This convergence justifies the transversal alignment activities described in clause 8.2 and the identification of the hENs requiring alignment presented in clause 8.3.5.

While the structural and functional analysis also shows that some hENs exhibit a level of convergence strong enough to justify evaluating a potential merging, the implications of the merger should not be considered only from the technical viewpoint but also from the regulatory and legal implications of potential mergers.

The present document thus provides a clear and technically grounded basis for the subsequent drafting of hENs to update them. The transversal alignment activities, the identification of affected hENs, together define the scope, priorities, and structural implications to be considered when updating the hENs.

10 Recommendations for selecting priority hENs for update

Further standardization updates of hENs should begin by integrating the aligned provisions identified in the present document, ensuring that architectural assumptions, antenna-related clauses, emission requirements, receiver performance aspects, and measurement methodologies are updated consistently across all affected hENs. This integration should reflect the unified operational assumptions and coherent treatment of dynamic pointing, scan-dependent gain, NGSO tracking, and other transversal elements documented in clause 8.2.

Future update of hENs should document the transversal elements and clause structures that should remain consistent in future revisions. This will support long-term maintenance of the hENs and ensure that future updates can be implemented efficiently and coherently.

The drafting sequence should follow a structured approach: first applying transversal alignment across all hENs identified then integrating alignment-related updates, and finally revising measurement annexes, definitions, and terminology. Cross-document validation should be performed to ensure that the updated hEN framework remains coherent and free of contradictions. Table 10-1 shows the hENs under review and the type of alignment activity.

Table 10-1: Harmonised Standards (hENs) under review

hEN number	hEN code	Title (Some of the titles need to be revised to align with RED and latest template)	Latest version revised	Alignment according to clause 8.3.5
First Priority				
1	ETSI EN 303 979 [i.4]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Earth Stations on Mobile Platforms (ESOMP) transmitting towards satellites in non-geostationary orbit, operating in the 27,5 GHz to 29,1 GHz and 29,5 GHz to 30,0 GHz frequency bands	V2.1.2	Group 1, Group 2, Group 4
2	ETSI EN 303 699 [i.2]	Satellite Earth Stations and Systems (SES); Fixed earth stations communicating with non-geostationary satellite systems in the 20 GHz and 30 GHz FSS bands; Harmonised Standard for access to radio spectrum	V1.2.0	Group 2, Group 3
3	ETSI EN 303 978 [i.3]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Earth Stations on Mobile Platforms (ESOMP) transmitting towards satellites in geostationary orbit, operating in the 27,5 GHz to 30,0 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.2.1	Group 1, Group 2, Group 4
4	ETSI EN 301 459 [i.10]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Satellite Interactive Terminals (SIT) and Satellite User Terminals (SUT) transmitting towards satellites in geostationary orbit, operating in the 29,5 GHz to 30,0 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.1.1	Group 2, Group 3
5	ETSI EN 301 427 [i.6]	Satellite Earth Stations and Systems (SES); Harmonised Standard for low data rate Mobile satellite Earth Stations (MES) except aeronautical mobile satellite earth stations, operating in the 11/12/14 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.1.25	Group 5
6	ETSI EN 302 448 [i.13]	Satellite Earth Stations and Systems (SES); Harmonised Standard for tracking Earth Stations on Trains (ESTs) operating in the 14/12 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.1.18	Group 1
7	ETSI EN 302 977 [i.16]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Vehicle-Mounted Earth Stations (VMES) operating in the 14/12 GHz frequency bands covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.1.13	Group 1, Group 4

hEN number	hEN code	Title (Some of the titles need to be revised to align with RED and latest template)	Latest version revised	Alignment according to clause 8.3.5
8	ETSI EN 302 340 [i.12]	Satellite Earth Stations and Systems (SES); Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 11/12/14 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU	V2.1.19	Group 2, Group 4
9	ETSI EN 301 443 [i.8]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Very Small Aperture Terminal (VSAT); Transmit only, transmit-and-receive, receive-only satellite earth stations operating in the 4 GHz and 6 GHz frequency bands	V2.2.0	Group 2, Group 3
10	ETSI EN 301 447 [i.9]	Satellite Earth Stations and Systems (SES); Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 4/6 GHz frequency bands allocated to the Fixed Satellite Service (FSS)	V2.1.12	Group 2, Group 4
11	ETSI EN 303 980 [i.5]	Satellite Earth Stations and Systems (SES); Fixed and in-motion Earth Stations communicating with non-geostationary satellite systems (NEST) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum	V1.3.1	Group 1, Group 3
12	ETSI EN 303 981 [i.17]	Satellite Earth Stations and Systems (SES); Fixed and in-motion Wide Band Earth Stations communicating with non-geostationary satellite systems (WBES) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum	V1.3.1	Group 1, Group 3
Second Priority				
13	ETSI EN 302 574-3 [i.14]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Mobile Earth Stations (MES) operating in the 1 980 MHz to 2 010 MHz (earth-to-space) and 2 170 MHz to 2 200 MHz (space-to-earth) frequency bands; Part 3: User Equipment (UE) for narrowband systems	V2.1.8	Group 5
14	ETSI EN 301 442 [i.7]	Satellite Earth Stations and Systems (SES); Harmonised Standard for NGSO Mobile Earth Stations (MES) including handheld earth stations, for Satellite Personal Communications Networks (S-PCN) operating in the 1 980 MHz to 2 010 MHz (earth-to-space) and 2 170 MHz to 2 200 MHz (space-to-earth) frequency bands under the Mobile Satellite Service (MSS)	V2.1.9	Group 5
15	ETSI EN 303 984 [i.15]	Satellite Earth Stations and Systems (SES); Earth Stations on board Aircraft (ESOA) communicating with non-geostationary satellite systems in the 10,7 GHz to 12,75 GHz and 12,75 GHz to 13,25 GHz frequency bands; Harmonised Standard for access to radio spectrum	V0.0.10 (2021)	Group 1, Group 4
16	ETSI EN 301 721 [i.11]	Satellite Earth Stations and Systems (SES); Harmonised Standard for Mobile Earth Stations (MES) providing Low Bit Rate Data Communications (LBRDC) using Low Earth Orbiting (LEO) satellites operating below 1 GHz frequency band	V2.2.0	Group 5

Annex A: Bibliography

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